Earth Tester



Category: **Electrical & Electronic Test Instruments** Type: **Full SOP** Priority: **High** 5 labs

Covers these lab items: Earth tester

What it is used for

An earth tester measures the resistance of an earthing system or electrode to help assess electrical safety. Correct test connections, suitable test conditions, and careful interpretation of readings support reliable results.

Main parts

- Display or scale shows measured earth resistance.
- Test terminals connect the instrument to the electrode and auxiliary test leads.
- Test leads connect the tester to the specified test points.
- Auxiliary spikes or probes provide reference points for applicable test methods.
- Selector switch chooses the test function or range.
- Test button initiates the measurement on applicable models.
- Battery or power source operates the instrument.

Before you start

1. Read the instrument manual and identify the approved earth-testing method.
2. Confirm authorisation and follow site electrical safety procedures before testing.
3. Inspect the tester, leads, clamps, and auxiliary spikes for damage.
4. Check the instrument's battery status and calibration or verification status.
5. Identify the earthing electrode and confirm whether the installation must be isolated for the selected test.
6. Select a suitable test location and check for buried services before inserting auxiliary spikes.

How to use

1. Switch off and isolate the relevant installation when required by the approved procedure.
2. Set up auxiliary spikes and test leads according to the selected measurement method.
3. Connect the tester terminals to the electrode and auxiliary leads as shown in the manual.
4. Check that all connections are secure and the leads are routed safely.
5. Select the correct test function and range.
6. Start the test using the designated control and wait for the reading to stabilise.
7. Record the resistance value, unit, test method, location, and relevant conditions.
8. Repeat the measurement if required by the procedure or if readings are unstable.
9. Disconnect the test leads and remove auxiliary spikes carefully.
10. Restore the installation only after the test area is clear and the approved procedure permits it.

⚠ Safety

- Only authorised personnel should test electrical earthing systems.
- Never disconnect protective earthing on an energised installation unless an approved safe procedure specifically permits it.
- Check for buried electrical cables and services before driving auxiliary spikes into the ground.
- Do not use damaged leads, probes, or instrument insulation.
- Keep test leads away from traffic, sharp edges, and wet areas where possible.
- Follow site isolation, lockout, and personal protective equipment requirements.

Care & storage

- Switch off the tester and disconnect leads before cleaning or storage.
- Clean the casing and test accessories with suitable dry materials.
- Inspect lead insulation, connectors, clamps, and spikes after use.
- Store the tester in its protective case and follow recommended calibration intervals.

Common mistakes to avoid

- Connecting test leads to the wrong terminals or test points.
- Ignoring buried services before placing auxiliary spikes.
- Testing an installation without required authorisation or isolation.
- Recording resistance values without units or the test method used.

Disclaimer: This SOP is prepared from general information available on the internet. For detailed instructions, refer to the manufacturer's original manual / file.

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